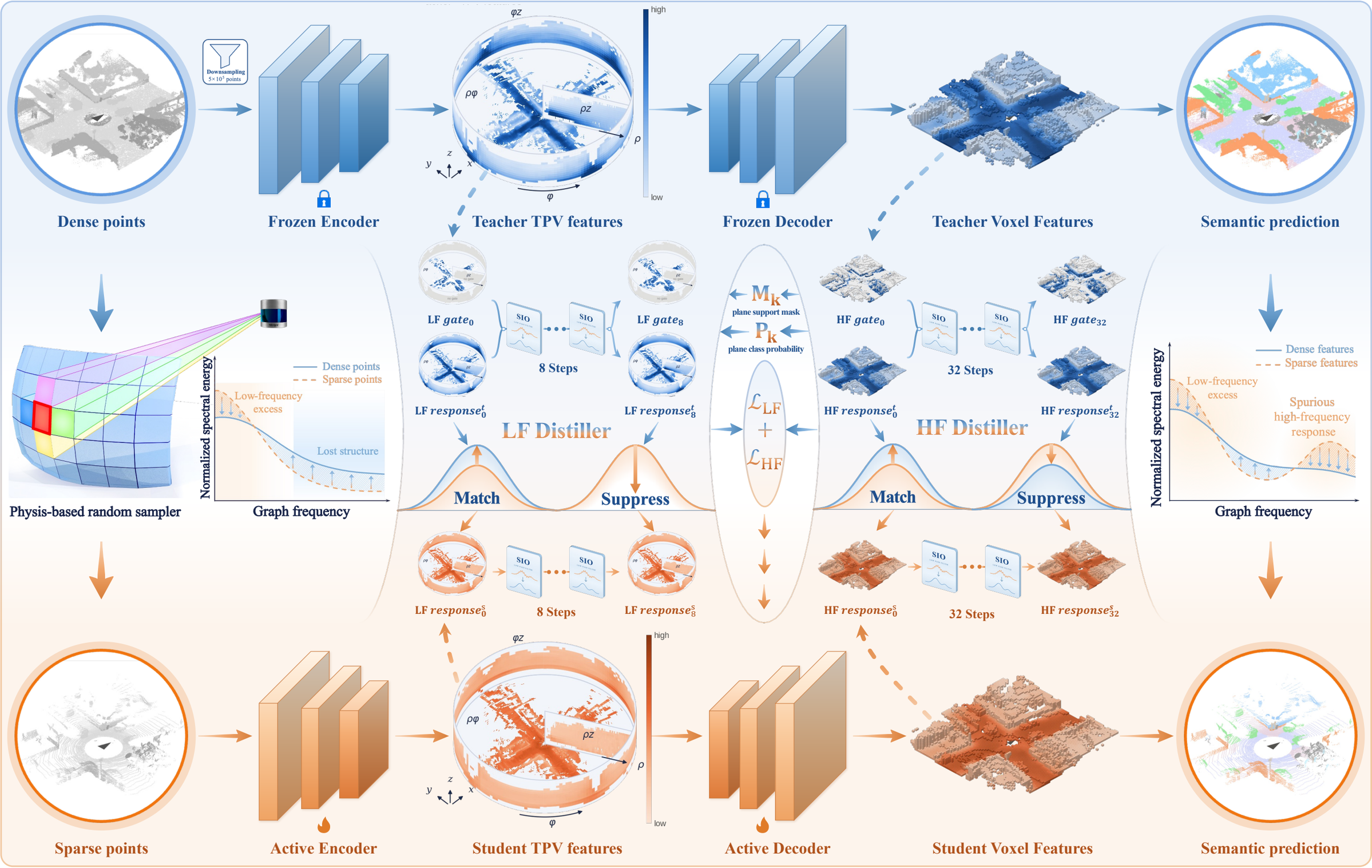

Dense points
Downsampling
5×10^5 points
Frozen Encoder
Teacher TPV features
high
low
Frozen Decoder
Teacher Voxel Features
Semantic prediction
LF $gate_0$
LF $gate_8$
SIO
8 Steps
LF $response_0^t$
LF $response_8^t$
M_k
plane support mask
P_k
plane class probability
$\mathcal{L}_{LF}$ + $\mathcal{L}_{HF}$
HF $gate_0$
HF $gate_{32}$
32 Steps
HF $response_0^t$
HF $response_{32}^t$
LF Distiller
HF Distiller
Match
Suppress
Physis-based random sampler
Normalized spectral energy
Dense points
Sparse points
Low-frequency excess
Lost structure
Graph frequency
Dense features
Sparse features
Spurious high-frequency response
LF $response_0^s$
LF $response_8^s$
HF $response_0^s$
HF $response_{32}^s$
Sparse points
Active Encoder
Student TPV features
Active Decoder
Student Voxel Features
Semantic prediction